

Reconsider case where break stick twice



$$U_i \sim \text{Unif}(0,1)$$

L = length of longest piece

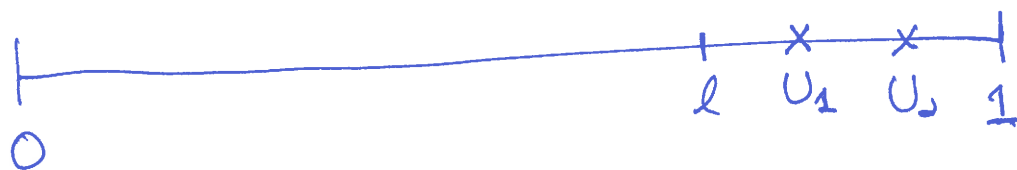
What is dist. of L ?

know $\frac{1}{3} \leq L \leq 1$, so $F_L(l) = \begin{cases} 0, & l < \frac{1}{3} \\ ?, & \frac{1}{3} \leq l \leq 1 \\ 1, & l > 1 \end{cases}$

Choose $\frac{1}{3} \leq l \leq 1$.

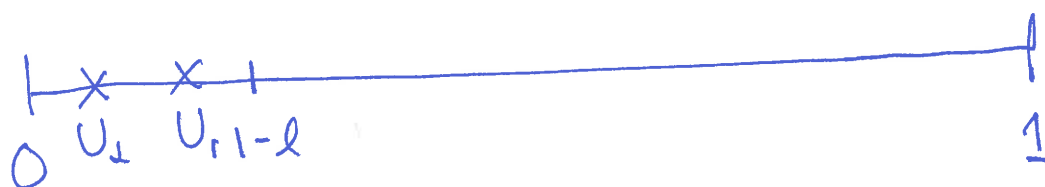
When $L \leq l$? When is $L > l$?

①



both U_1 and $U_2 > l$

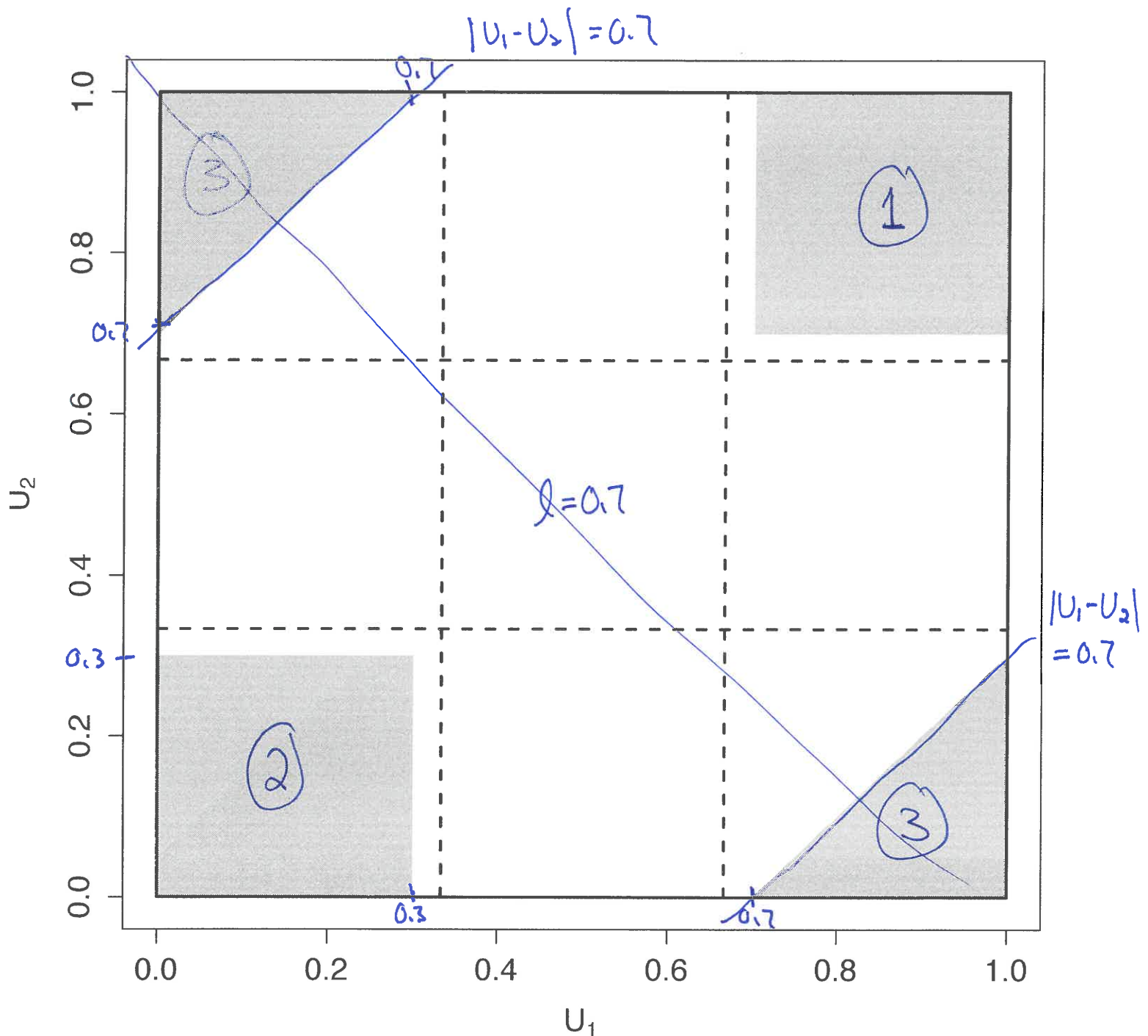
②



both U_1, U_2 less than $1-l$

③ When $|U_1 - U_2| > \ell$

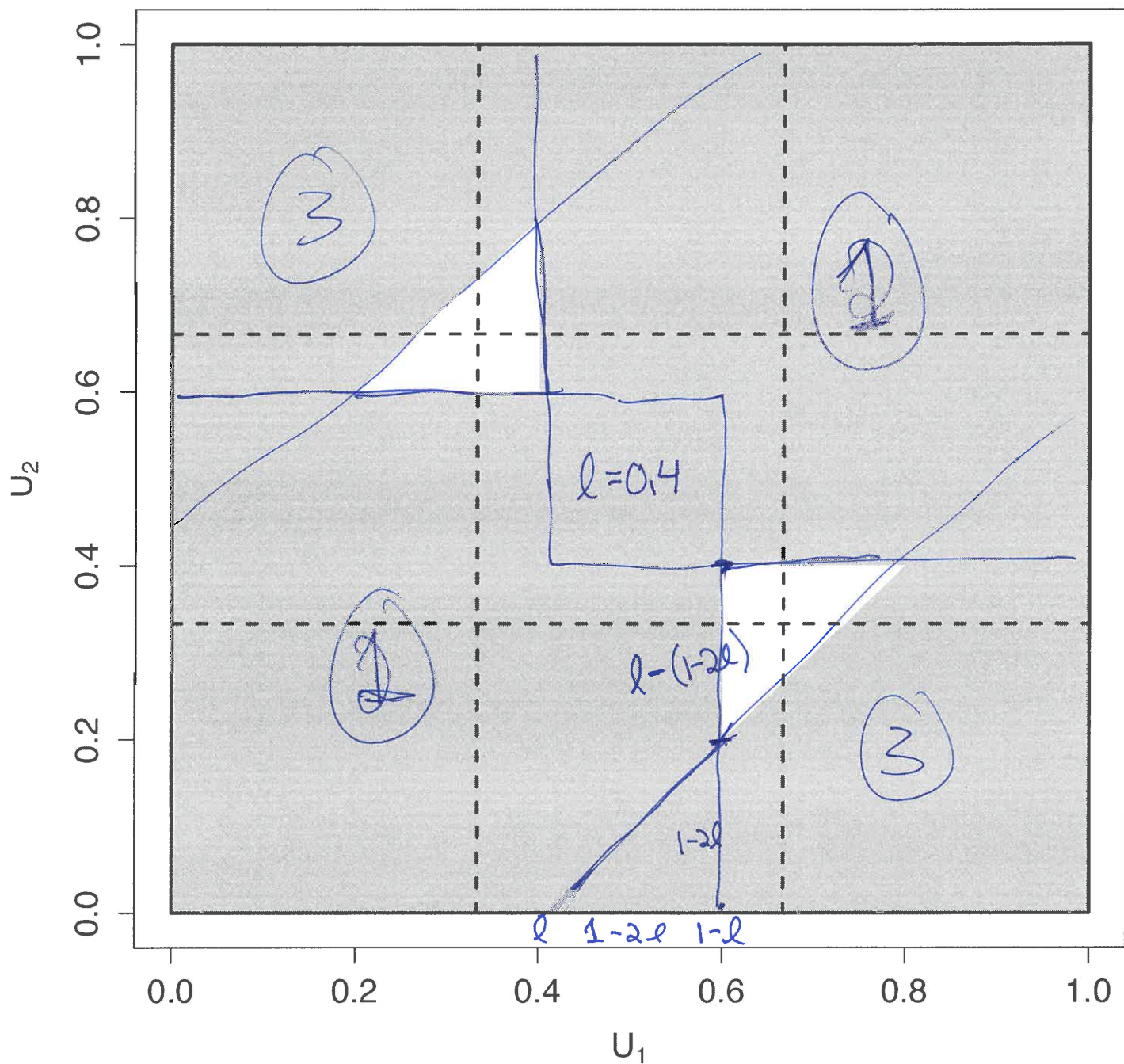




Case where $0.5 \leq l \leq 1$

$$P(L > l) = \text{area of shaded regions} \\ = 3(1-l)^2$$

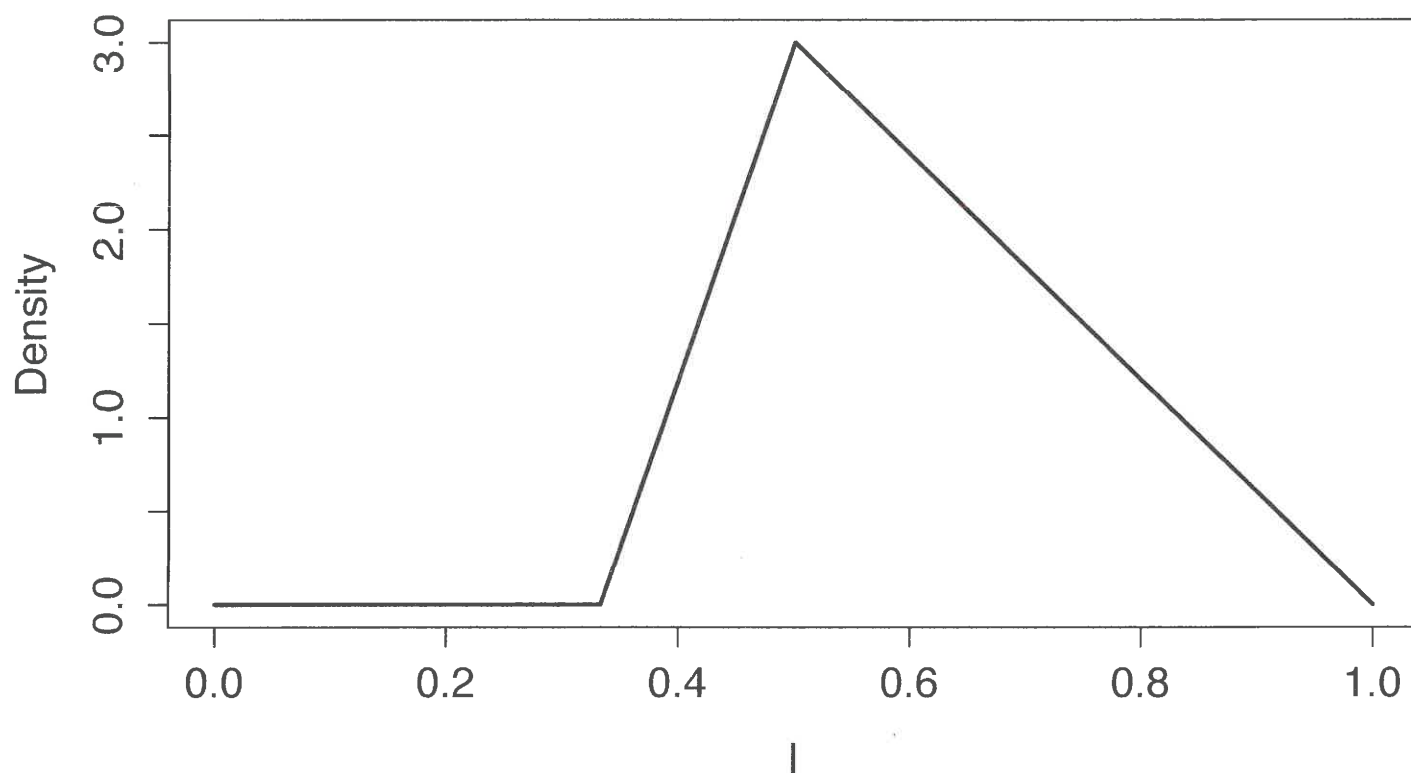
$$P(L \leq l) = 1 - 3(1-l)^2$$



Case $\frac{1}{3} \leq l \leq \frac{1}{2}$

$$P(L \leq l) = \text{area of unshaded region} \\ = (3l-1)^2$$

$$F_L(l) = \begin{cases} 0, & l < \frac{1}{3} \\ (3l-1)^2, & \frac{1}{3} \leq l < \frac{1}{2} \\ 1-3(1-l)^2, & \frac{1}{2} \leq l \leq 1 \\ 1, & l > 1 \end{cases}$$



$$f_L(l) = \begin{cases} 6(3l-1), & \frac{1}{3} \leq l < \frac{1}{2} \\ 6(1-l), & \frac{1}{2} \leq l \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$P(\text{form a } \Delta) = F_L\left(\frac{1}{2}\right) = \frac{1}{4}$$

1 The Pareto(α, λ) Distribution

2 The random variable X has the **Pareto(α, λ) distribution** if $\alpha > 0$ and $\lambda > 0$
3 with

$$4 \quad f_X(x) = \begin{cases} \frac{\alpha \lambda^\alpha}{x^{\alpha+1}}, & x \geq \lambda \\ 0, & \text{otherwise} \end{cases}.$$

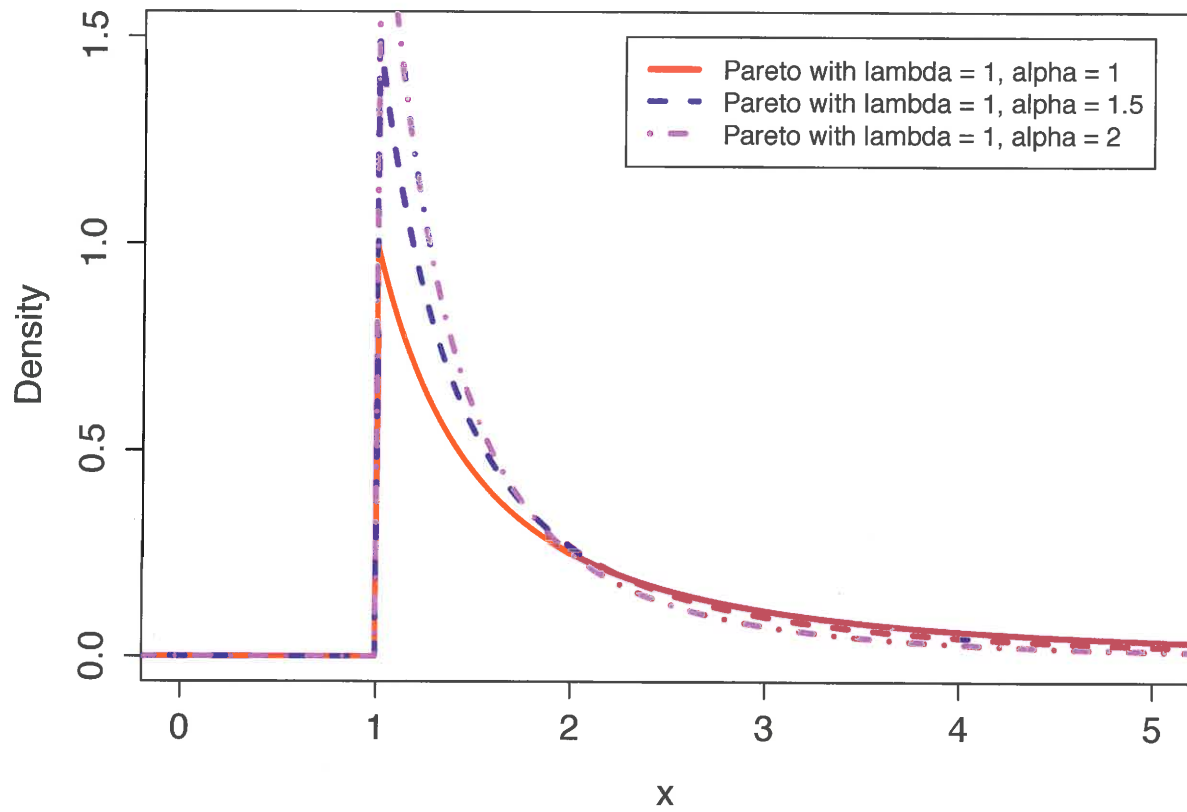
5 This is often used for modelling the tail of a distribution.

6 Some useful properties:

$$7 \quad E(X) = \begin{cases} \infty, & \alpha \leq 1 \\ \frac{\alpha \lambda}{\alpha - 1}, & \alpha > 1 \end{cases}$$

$$8 \quad V(X) = \begin{cases} \text{undefined}, & \alpha \leq 1 \\ \infty, & 1 < \alpha \leq 2 \\ \frac{\alpha \lambda}{(\alpha - 1)^2 (\alpha - 2)}, & \alpha > 2 \end{cases}$$

$$9 \quad F_X(x) = \begin{cases} 1 - \left(\frac{\lambda}{x}\right)^\alpha, & x \geq \lambda \\ 0, & x < \lambda \end{cases}$$



The Exponential(λ) Distribution

When X has the **Exponential(λ) distribution**, the following hold:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = 1/\lambda \text{ and } V(X) = 1/\lambda^2$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

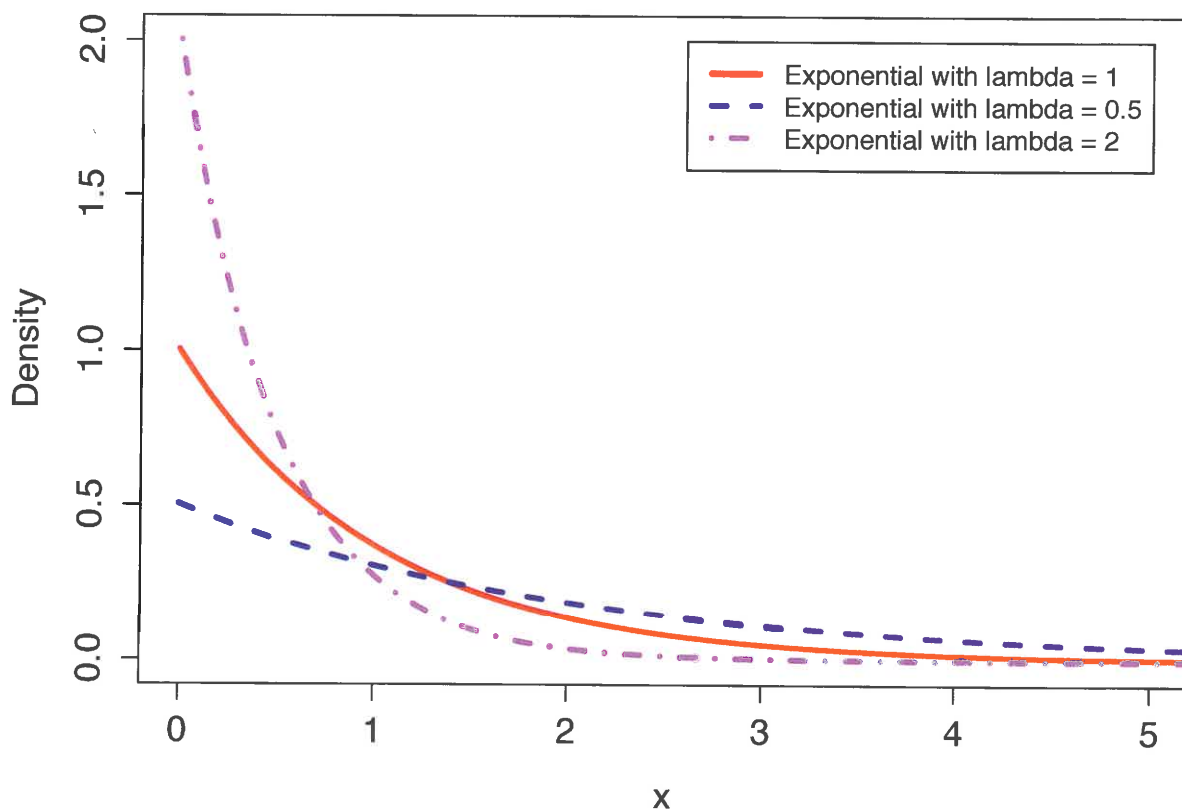
← "rate parameterization"

$f_X(x) = \frac{1}{\beta} e^{-x/\beta}$

$E(X) = \beta$

$V(X) = \beta^2$ ↑

"mean parameterization"



- 1 **Exercise:** Suppose that X has the Exponential(λ) distribution, and that
2 $a > 0$ and $b > 0$. Show that

$$P(X > a + b | X > b) = P(X > a).$$

- 4 Why is this called the "Memoryless Property?"

$$\begin{aligned} P(X > a + b | X > b) &= \frac{P(X > a + b \text{ and } X > b)}{P(X > b)} \\ &= \frac{P(X > a + b)}{P(X > b)} \\ &= \frac{e^{-\lambda(a+b)}}{e^{-\lambda b}} = e^{-\lambda a} = P(X > a) \end{aligned}$$

14 e.g. If I model the length of life for a
15 lightbulb as exponential(λ), the
16 prob. the bulb lasts 10 more days
17 is always the same, no matter how
18 old it is.

The Gamma(α, β) Distribution

For $\alpha > 0$ and $\beta > 0$, random variable X has the **Gamma(α, β) distribution** when

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1} e^{-\beta x} \beta^\alpha}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

where

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \text{ is the Gamma Function.}$$

Properties worth noting: If $X \sim \text{Ga}(\alpha, \beta)$, then

1. $E(X) = \alpha/\beta$

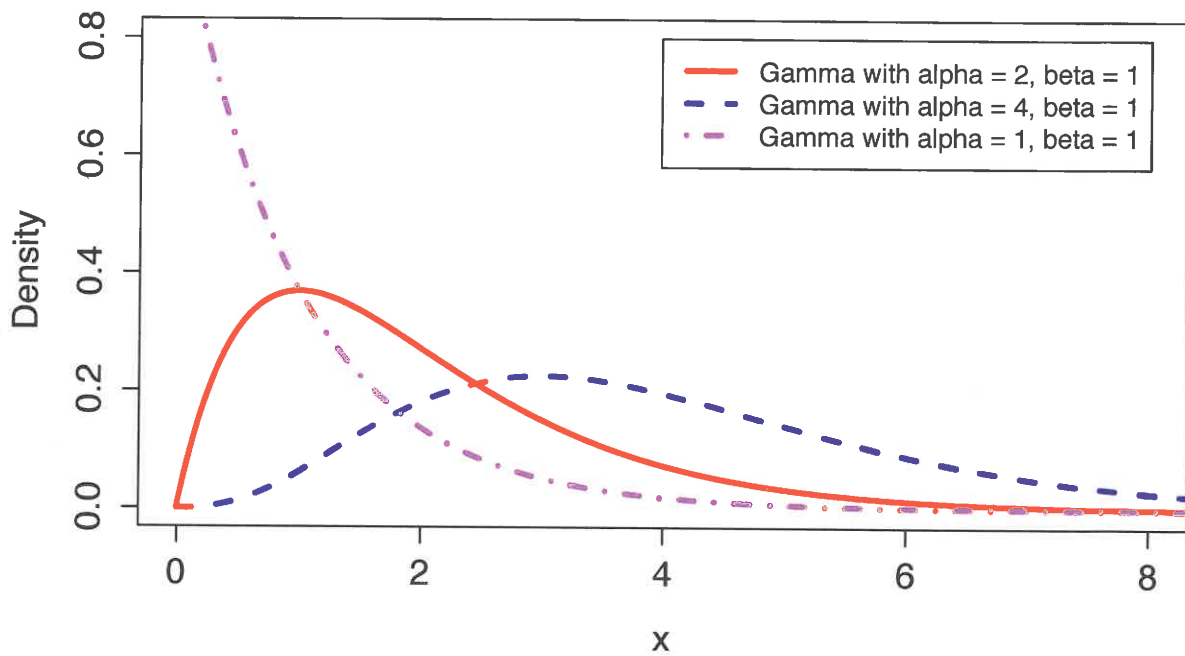
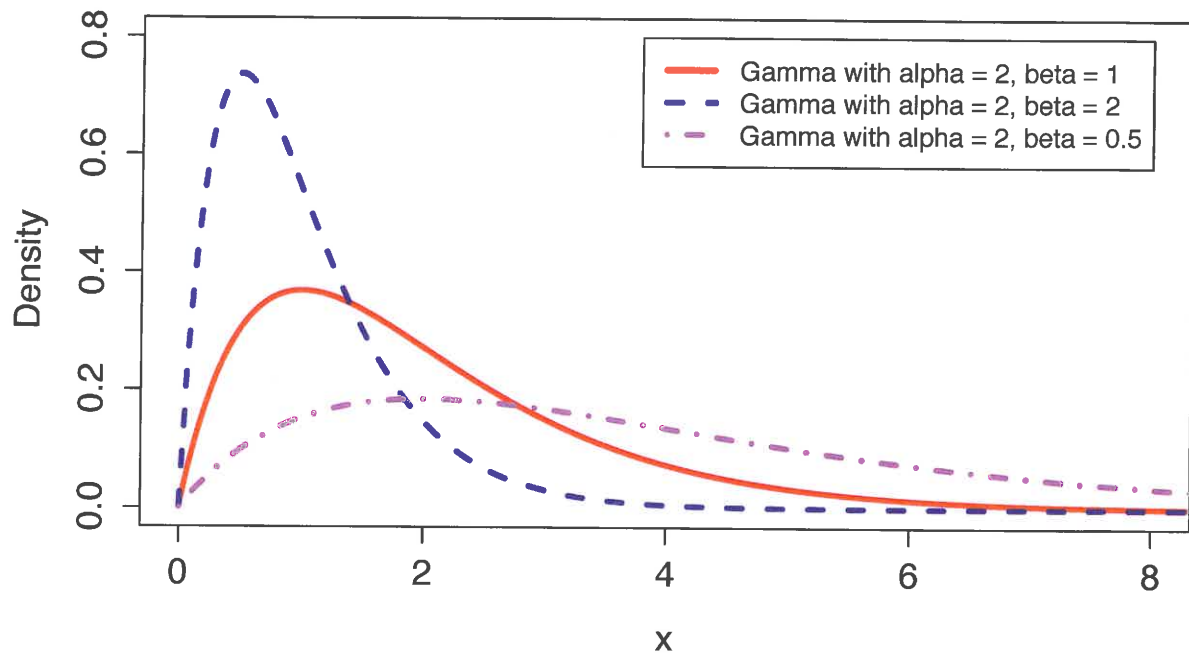
2. $V(X) = \alpha/\beta^2$

3. If $\alpha = 1$, then $X \sim \text{Exp}(\beta)$.

4. For any $c > 0$, the random variable $cX \sim \text{Ga}(\alpha, \beta/c)$.

5. If $X_1, X_2, \dots, X_n$ are independent random variables with $X_i \sim \text{Ga}(\alpha_i, \beta)$, then $\sum_i X_i \sim \text{Ga}(\sum_i \alpha_i, \beta)$.

6. The sum of n independent $\text{Exponential}(\beta)$ random variables has the **Gamma(n, β) distribution**. This is called the **Erlang Distribution**.



- 1 **Exercise:** You are waiting in line at the express checkout of the grocery
 2 store. Three customers are waiting ahead of you and one customer is cur-
 3 rently being served. The checkout time for an individual customer has an
 4 exponential distribution with mean of 20 seconds. What is ^{the} probability that
 5 you will still be waiting for service to begin two minutes from now?

6 Note that the exponential assumption allows me
 7 to calculate this prob. without knowing
 8 how long it has been since the person
 9 being served started service

$$10 \quad T = \text{time until I start being served (in minutes)}$$

$$11 \quad = T_0 + T_1 + T_2 + T_3$$

13 T_0 = remaining service time for cust. being served

14 T_1 = service time for person first in line

15 T_2 = " " " " second "

16 T_3 = " " " " third "

$$18 \quad T_0, T_1, T_2, T_3 \sim \text{Exponential}(\lambda=3)$$

20 If I assume RVs are independent,
 21 then $T \sim \text{Ga}(4, 3)$

1 The Normal(μ, σ^2) Distribution

2 The random variable X has the **Normal(μ, σ^2) Distribution** if, for $\sigma > 0$,

3
$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad -\infty < x < \infty$$

4 This is sometimes called the **Gaussian Distribution**.

5 The case where $\mu = 0$ and $\sigma = 1$ is called the **standard normal distribution**.

6 The CDF of the standard normal distribution is commonly denoted with
7 Φ .

8 **Properties worth noting:** If $X \sim N(\mu, \sigma^2)$, then

9 1. $E(X) = \mu$

10 2. $V(X) = \sigma^2$

11 3. For $a \neq 0$ and any b , the random variable $aX + b$ has the $N(a\mu + b, a^2\sigma^2)$
12 distribution. Note that this implies that $(X - \mu)/\sigma$ has the standard
13 normal distribution.

$$a = 1/\sigma \quad b = -\mu/\sigma$$

14 4. Linear combinations of independent normal random variables are also
15 normally distributed.

Exercise: Suppose that $\log(S)$ has the $\text{Normal}(\mu, \sigma^2)$ distribution. Let K be a positive constant. What is $E((S - K)^+)$?

Note: This can be thought of as the expected value of a call option with strike price K , with S representing the price of the underlying stock on the maturity date.

$$Y^+ = \begin{cases} Y, & Y > 0 \\ 0, & Y \leq 0 \end{cases}$$

$$Y = \log(S) \sim N(\mu, \sigma^2)$$

$$S \text{ is lognormal}(\mu, \sigma^2)$$

$$E((S - K)^+) = E((e^Y - K)^+)$$

$$= \int_{-\infty}^{\infty} (e^y - K)^+ f_Y(y) dy \quad \begin{array}{l} \text{Rule for} \\ \text{Lazy Stat.} \end{array}$$

when is $e^y < K$? when $y < \log(K)$

$$= \int_{\log(K)}^{\infty} (e^y - K) f_Y(y) dy$$

$$= \int_{\log K}^{\infty} e^y f_Y(y) dy - \int_{\log K}^{\infty} K f_Y(y) dy$$

$$\int_{\log K}^{\infty} e^y f_Y(y) dy = \int_{\log K}^{\infty} e^y \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right) dy$$

$$= \int_{\log k}^{\infty} e^y \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\left(\frac{y^2 - 2\mu y + \mu^2}{2\sigma^2}\right)\right) dy$$

$$= \int_{\log k}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\left(\frac{y^2 - 2\mu y + \mu^2 - 2\sigma^2 y}{2\sigma^2}\right)\right) dy$$

$$= \int_{\log k}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\left(\frac{y^2 - 2y(\mu + \sigma^2) + \mu^2}{2\sigma^2}\right)\right) dy$$

$$= \int_{\log k}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\left(\frac{y^2 - 2y(\mu + \sigma^2) + (\mu + \sigma^2)^2 - (\mu + \sigma^2)^2 + \mu^2}{2\sigma^2}\right)\right) dy$$

$$= \int_{\log k}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y - (\mu + \sigma^2))^2}{2\sigma^2}\right) \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) dy$$

$$= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) P(X > \log k)$$

$$\text{where } X \sim N(\mu + \sigma^2, \sigma^2)$$

$$= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) P\left(\frac{X - \mu - \sigma^2}{\sigma} > \frac{\log k - \mu - \sigma^2}{\sigma}\right)$$

$$= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) \left(1 - \Phi\left(\frac{\log k - \mu - \sigma^2}{\sigma}\right)\right)$$

$$= \exp\left(\frac{(\mu + \sigma^2)^2 - \mu^2}{2\sigma^2}\right) \Phi\left(\frac{(\mu + \sigma^2) - \log k}{\sigma}\right)$$

$$\Phi(x) = 1 - \Phi(-x)$$

$$= \exp\left(\mu + \frac{\sigma^2}{2}\right) \Phi\left(\frac{(\mu + \sigma^2) - \log k}{\sigma}\right)$$

$$\begin{aligned} k \int_{\log k}^{\infty} f_Y(y) dy &= k P(Y > \log k) \\ &= k P\left(\frac{Y - \mu}{\sigma} > \frac{\log k - \mu}{\sigma}\right) \\ &= k \Phi\left(\frac{\mu - \log k}{\sigma}\right) \end{aligned}$$

$$\begin{aligned} E((S - k)^+) &= \exp\left(\mu + \frac{\sigma^2}{2}\right) \Phi\left(\frac{(\mu + \sigma^2) - \log k}{\sigma}\right) \\ &\quad - k \Phi\left(\frac{\mu - \log k}{\sigma}\right) \end{aligned}$$

The Poisson Process

In our initial discussion of the $\text{Poisson}(\lambda)$ distribution, we noted how the Poisson can be derived from the limit of $\text{binomial}(n, \lambda/n)$ random variables. There we imagined the interval $(0, 1)$ divided into n subintervals.



Now, imagine the same construction, but with the time interval extended to indefinitely. This leads to the **Poisson Process with rate λ** .

If I say that “events occur as a Poisson Process with rate λ ,” then

1. If Y equals the number of events during an interval of time of length T , then Y has the $\text{Poisson}(\lambda T)$ distribution.

2. The times between events are random variables with the $\text{Exponential}(\lambda)$ distribution.

3. The times between events are independent random variables.

4. The numbers of events in disjoint time intervals are independent random variables.

5. The waiting time for k events has the $\text{Gamma}(k, \lambda)$ distribution. (This is also called the Erlang distribution.)

Exercise: Show that Property (1) above implies Property (2).

Suppose that time 0 corresponds to the occurrence of an event

T = time of subsequent event

For $t > 0$,

$$F_T(t) = P(T \leq t) = 1 - P(\text{at least one event during } (0, t))$$

$$= 1 - P(\text{no events during } (0, t))$$

$$= 1 - \frac{e^{-\lambda t} (\lambda t)^0}{0!}$$

since # of events
during $(0, t) \sim \text{Poisson}(\lambda t)$

$$= 1 - e^{-\lambda t}$$

So, by uniqueness of CDF, $T \sim \text{Exp}(\lambda)$

Exercise: Suppose that X_2 equals the number of “jumps” between now and maturity date for a call option. Assume that these jumps arrive as a Poisson process with rate λ (in units of day^{-1}). When a jump does occur, its size is normally distributed with mean zero and variance τ^2 . There is still an underlying normal distribution to model the “normal” fluctuations in the stock price on the log-scale. Let T denote the time to maturity (in days).

So, the model can be written as follows: If S is the price on the maturity date, then

$$S = \exp \left(X_1 + \sum_{i=1}^{X_2} Z_i \right)$$



where X_1 has the $\text{Normal}(\mu, \sigma^2)$ distribution, X_2 has the $\text{Poisson}(\lambda T)$ distribution, and the Z_i have the $\text{Normal}(0, \tau^2)$ distribution.

Let K be the strike price for the option. What is $E((S - K)^+)$? You can assume that all of the random variables are independent.

$$E((S - K)^+) = \sum_{j=0}^{\infty} E((S - K)^+ | X_2 = j) P(X_2 = j)$$

a version of LFTP for expectations

$$\text{when } X_2 = j, \quad X_1 + \sum_{i=1}^{X_2} Z_i \sim N(\mu, \sigma^2 + j\tau^2)$$

So, $E((S-k)^+ | X_2=j)$ can be

found using the previous "Options" exercise. Just note that the variance of S is $\sigma^2 + j\tau^2$ instead of σ^2

$$P(X_2=j) = \frac{e^{-\lambda\tau} (\lambda\tau)^j}{j!}$$

Since $X_2 \sim \text{Poisson}(\lambda\tau)$

Properties of Expectation

If X, Y are random variables, and a, b , and c are constants, then

1. $E(c) = c$

2. $E(cX) = cE(X)$

3. $E(X + Y) = E(X) + E(Y)$ if either

(a) X, Y are both nonnegative

(b) X, Y are both nonpositive

(c) At least one of X, Y has finite expected values

4. $E(aX + b) = aE(X) + b$

5. $V(aX + b) = a^2V(X)$

$$E(X) = \infty$$
$$Y = -X$$

Jensen's Inequality

Property (4) on the previous page shows that if $g(x) = ax + b$, then $E(g(X)) = g(E(X))$. Unfortunately, however, it is not generally true that $E(g(X)) = g(E(X))$. What can be said?

Jensen's Inequality gives that if $g(\cdot)$ is convex on the support of the random variable X , then

$$E(g(X)) \geq g(E(X)).$$

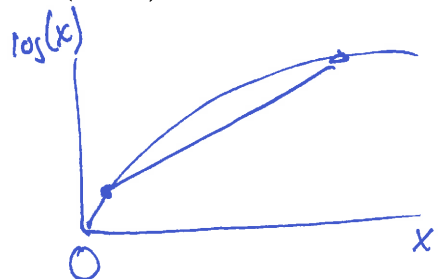
Likewise, if $g(\cdot)$ is concave on the support of the random variable X , then

$$E(g(X)) \leq g(E(X)).$$



Example 1: The function $g(x) = x^2$ is convex on the entire real line, so for any random variable X , $E(X^2) \geq E(X)^2$. (Note that $V(X) = E(X^2) - E(X)^2$.)

Example 2: The function $g(x) = \log(x)$ is concave on $(0, \infty)$, so if X is a positive random variable, $E(\log(X)) \leq \log(E(X))$.



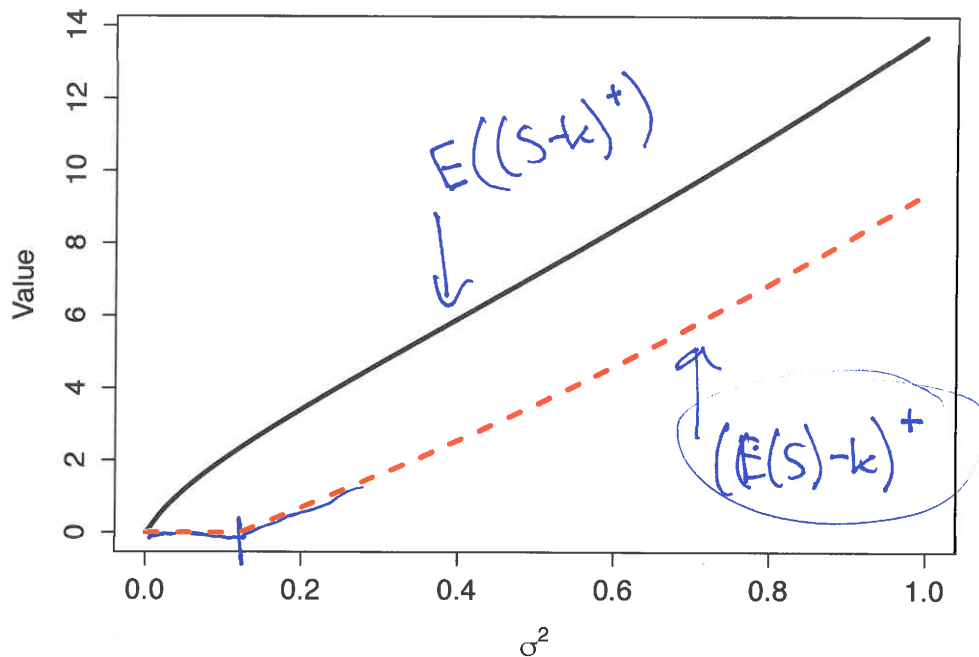
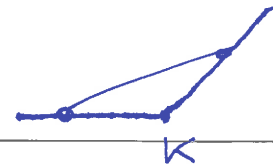


Figure 1: The case where $\mu = \log(16)$ and $K = 17$. One line shows the value of $E((S - K)^+)$, while the other is $(E(S) - K)^+$.

Exercise: Reconsider the exercise above in which $\log(S)$ was assumed to be $N(\mu, \sigma^2)$ and K is a positive constant. Figure 1 shows a case where $\mu = \log(16)$ and $K = 17$. The value of σ^2 varies along the horizontal axis. One line shows the value of $E((S - K)^+)$, while the other is $(E(S) - K)^+$. Which one is which?

$$g(x) = (x - k)^+ \text{ is convex}$$



$$E(g(S)) \geq g(E(S))$$

$$E((S - k)^+) \geq (E(S) - k)^+$$

Part 3: Multivariate Distributions

Exercise: Suppose that X and Y have pmf's

$$f_X(x) = \begin{cases} 0.5, & x = 100 \\ 0.5, & x = 250 \end{cases}$$

and

$$f_Y(y) = \begin{cases} 0.25, & y = 0 \\ 0.25, & y = 100 \\ 0.5, & y = 200 \end{cases}$$

What is $P(X > Y)$?

$$\begin{aligned} P(X > Y) &= P(X > Y | Y=0) P(Y=0) \\ &\quad + P(X > Y | Y=100) P(Y=100) \\ &\quad + P(X > Y | Y=200) P(Y=200) \quad \text{LFTP} \end{aligned}$$

If I assume $P(X > Y | Y=100) = 0.5 = P(X=250)$
then are assuming no dependence between
 X and Y . A strong assumption.

As stated, there is not enough information
to answer the question

1 In the case of a pair of discrete random variables X and Y , we need to
2 specify the **joint probability mass function** in order questions such as the
3 previous one. We naturally write

$$4 \quad f_{X,Y}(x, y) = P(X = x, Y = y)$$

5 Note that $f_{X,Y}(x, y) \geq 0$ for all x and y and that

$$6 \quad \sum_x \sum_y f_{X,Y}(x, y) = 1$$

7 where the sum is over all pairs (x, y) for which $f_{X,Y}(x, y) > 0$.

8 The CDF is defined as

$$9 \quad F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

10 The extension to more than two random variables is simple.

- 1 **Exercise:** A large insurance agency services a number of customers who
2 have purchased both a homeowner's policy and an automobile policy from
3 the agency. For each type of policy, a deductible amount must be specified.
4 Let X be the deductible amount on the auto policy, and Y the deductible
5 amount on the homeowner's policy. The joint pmf of (X, Y) is given by
6 the following table:

x	y		
	0	100	200
100	.20	.10	.20
250	.05	.15	.30

- 8 What is the probability that X is larger than Y ?

$$P(X > Y) = \sum_{\substack{(x,y): \\ x > y}} f_{X,Y}(x,y)$$

$$= f_{X,Y}(100, 0) + f_{X,Y}(250, 0) \\ + f_{X,Y}(250, 100) + f_{X,Y}(250, 200)$$

$$= 0.70$$

The Continuous Case

The pair of random variables (X, Y) is **jointly continuous** with density $f_{X,Y}(x, y)$ if

1. $f_{X,Y}(x, y) \geq 0$ for all x, y

2.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$$

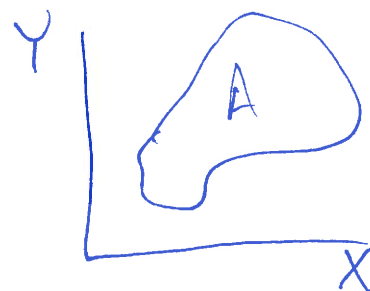
To calculate probabilities, integrate the pdf over the region of interest:

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

Here, it is **not** the case that A need be rectangular.

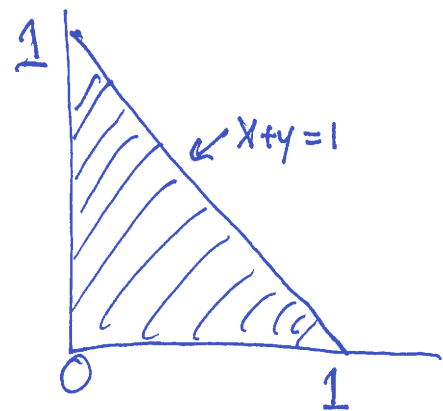
A useful property: In the continuous case,

$$\frac{\partial F_{X,Y}(x, y)}{\partial x \partial y} = f_{X,Y}(x, y)$$



Exercise: Suppose that the joint pdf of (X, Y) is given by:

$$f_{X,Y}(x, y) = \begin{cases} 24xy, & x > 0, y > 0 \\ & x + y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



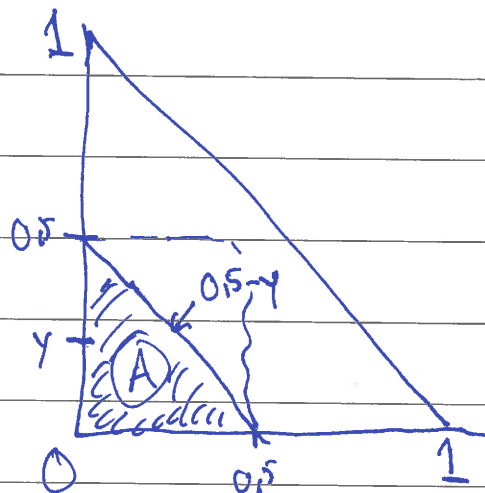
1. What is the probability that $X + Y$ is at most 0.5?

2. Find $P(X > Y)$.

3. Find the pdf of the random variable $T = X + Y$.

① $P(X + Y \leq 0.5)$

$P((X, Y) \in A)$



$$= \iint_A f_{X,Y}(x, y) dx dy = \int_0^{0.5} \int_0^{0.5-y} 24xy dx dy$$

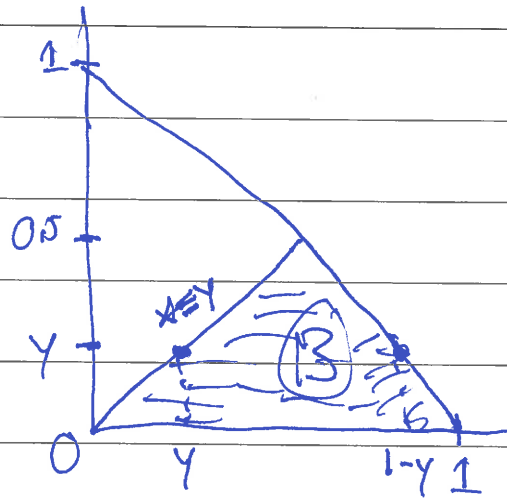
$$= \frac{1}{16}$$

$$(2) P(X > Y)$$

$$= P((X, Y) \in B)$$

$$= \int_0^{0.5} \int_y^{1-y} 24xy \, dx \, dy$$

$$= 0.5$$



$$(3) \text{ pdf of } T = X + Y$$

Find $F_T(t)$, take derivative

for $0 < t < 1$,

$$P(T \leq t) = P(X + Y \leq t)$$

$$= \int_{y=0}^t \int_{x=0}^{t-y} 24xy \, dx \, dy = t^4$$

So,

$$F_T(t) = \begin{cases} 0, & t < 0 \\ t^4, & 0 \leq t \leq 1 \\ 1, & t > 1 \end{cases}$$

So,

$$f_T(t) = \begin{cases} 4t^3, & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases}$$

Don't write $f_T(t) = 4t^3$

Ok to write $f_T(t) = 4t^3, 0 < t < 1$